PRACTICAL 4

Setting Up a Local Area Network (LAN)

1. Aim

To design, set up, and configure a Local Area Network (LAN) connecting four computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, and demonstrate resource/data sharing across the network.

2. Equipment Required

S.No.	Equipment	Quantity	Specification
1	Personal Computers (PC)	4	Ethernet (NIC) enabled
2	Network Switch	1	8 / 16 / 24 port, unmanaged/managed
3	Ethernet Cables (Cat5e/Cat6)	4	Straight-through, RJ-45 connectors

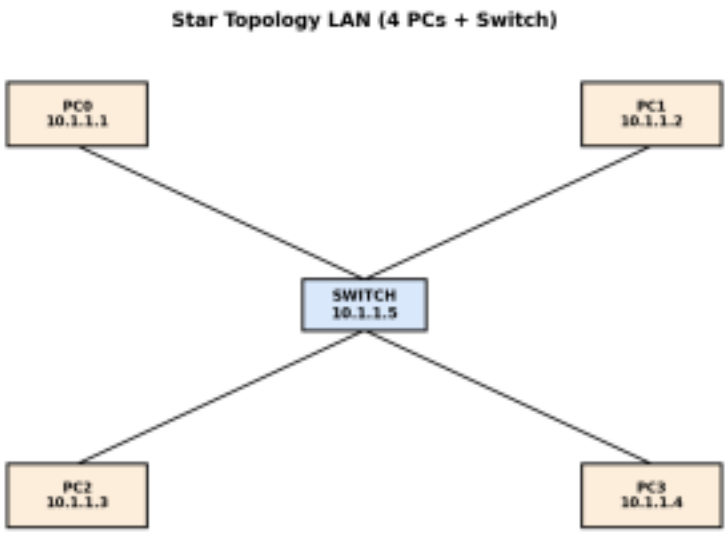
3. Theory

A Local Area Network (LAN) is a network that connects computers and devices within a limited geographical area such as a home, office, or laboratory, allowing them to share resources such as files, printers, and internet connections.

A Star Topology is used in this experiment, where all end devices (PCs) are individually connected to a central network switch. The switch operates at Layer 2 of the OSI model and forwards data frames between connected devices based on their MAC addresses. Every device on a LAN must be assigned an IP address within the same network range (as determined by the subnet mask) to be able to communicate with one another.

4. Network Diagram

The LAN was implemented using a Star Topology as shown below, with 4 PCs connected individually to a central switch.



5. IP Address Configuration Table

Device	IP Address	Subnet Mask	Connected To
PC0	10.1.1.1	255.0.0.0	Switch Port Fa0/1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa0/2
PC2	10.1.1.3	255.0.0.0	Switch Port Fa0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa0/4
Switch (Management)	10.1.1.5	255.0.0.0	VLAN 1 Interface

6. Implementation in Cisco Packet Tracer

Since physical hardware was limited, the LAN was also implemented and verified using Cisco Packet Tracer simulation software, following the steps below.

Step 1: Place the Devices

1. Open Cisco Packet Tracer.
2. From the device category panel, select End Devices and drag 4 PCs onto the workspace.
3. Select Switches and drag 1 switch (e.g., 2960 series) onto the workspace.

Step 2: Cable the Devices

4. Click Connections → select Copper Straight-Through cable.
5. Click each PC, then click the switch, and select an available FastEthernet port to complete each connection.
6. Repeat for all four PCs (ports Fa0/1 to Fa0/4). Green link lights confirm a successful physical connection.

Step 3: Assign IP Addresses to PCs

7. Click a PC → Desktop tab → IP Configuration.

8. Enter the Static IP Address and Subnet Mask as per the IP table (Section 5) for each PC.

Step 4: Configure the Switch Management IP (Optional)

9. Click the switch → CLI tab.
10. Enter the following commands to assign a management IP:

```
enable
configure terminal
interface vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit
```

Step 5: Verify Connectivity Using Ping

11. Click a PC → Desktop tab → Command Prompt.
12. Use the ping command to test connectivity to the other PCs, e.g., ping 10.1.1.2.
13. Successful “Reply from...” messages confirm connectivity.

Step 6: Demonstrate Data Sharing (Server Simulation)

14. Add a Server device (End Devices → Server-PT) and connect it to the switch.
15. Assign it an IP address (e.g., 10.1.1.10, subnet mask 255.0.0.0) via its Desktop → IP Configuration tab.
16. Open the server's Services (Config) tab and enable the HTTP service.
17. From another PC, open the Web Browser (Desktop tab) and enter the server's IP address to load the page, confirming successful data transfer across the LAN.

Step 7: Observe Packet Flow in Simulation Mode

18. Switch to Simulation mode (bottom-right of the workspace).
19. Send a ping or web request between two devices and use the Play/Capture controls to step through the packet as it travels hop-by-hop through the switch, observing MAC-address-based forwarding.

7. Observation

All four PCs were successfully connected to the switch in a star topology and assigned unique static IP addresses within the same subnet (10.0.0.0/255.0.0.0). Ping tests between all devices returned successful replies, confirming end-to-end connectivity. The shared folder/server resource was successfully accessed from other computers on the network, confirming that the LAN was functioning correctly for resource sharing.